

RAY OPTICS

Ray optics (or geometrical optics) describes light propagation in terms of *rays*. It can be used to describe phenomena like reflection and refraction of light and is therefore useful for the understanding of optical devices like mirrors or lenses. It does, however, not take into account wave aspects of light, such as diffraction and interference, which are explained by *wave optics*.

LIGHT

Light can be described as an *electromagnetic wave*. For monochromatic light (i.e. light with a single spectral colour), the electric and magnetic wave are given by

$$E(x, t) = E_0 \cdot \cos(2\pi f \cdot t - \frac{2\pi}{\lambda} \cdot x) \quad \text{and} \quad B(x, t) = B_0 \cdot \cos(2\pi f \cdot t - \frac{2\pi}{\lambda} \cdot x)$$

where f and λ are the *frequency* and *wavelength*. They are related to the propagation speed through the relation

$$c = \lambda f$$

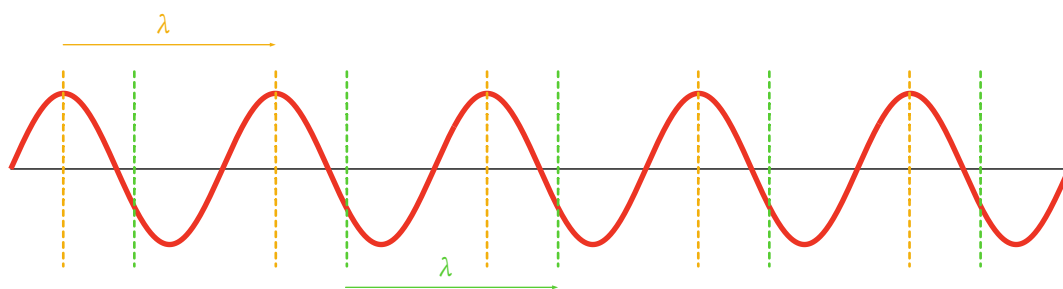
The propagation speed in a vacuum is the same for any electromagnetic wave: $c = c_{\text{vac}} = 2.99792458 \cdot 10^8 \text{ m/s}$.

WAVEFRONT AND RAY

The *phase* of a wave is the argument

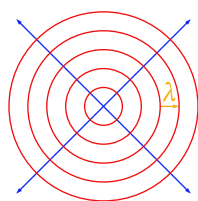
$$\varphi(t) = 2\pi a f \cdot t - \frac{2\pi}{\lambda} \cdot x$$

Points with the same phase define a *wavefront*. In one dimension, wavefronts correspond to single points, in two dimensions to curves, and in three dimensions to surfaces. Waves are often visualised by drawing wavefronts that correspond to a certain displacement, e.g. to wave maxima. The distance between two neighbouring wave fronts is then equal to the wavelength.

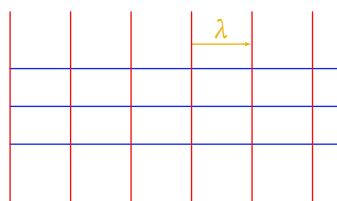


A *light ray* is a line or curve that is perpendicular to the wavefronts. This is equivalent to *Fermat's principle*, which states that the path taken between two points by a ray of light is the path that can be traversed in the least time.

The two most basic wave types are circular/spherical waves (in two/three dimensions) and plane waves.



circular wave



plane wave

wavefront
light ray

LIGHT IN A MEDIUM

As mentioned above, all electromagnetic waves travel at the speed of light when propagate in a vacuum. As a consequence, waves are spherical waves, or – far away from the wave source – plane waves. In each case, the rays are straight lines.

The propagation speed in a medium (e.g. water or glass) is smaller than in a vacuum. In a *homogeneous* medium, the speed does not depend on the position in the medium. In an *isotropic* medium, the propagation does not depend on the direction.

REMARKS:

- Examples of *inhomogeneous* media are gases with a temperature gradient (e.g. the air over a hot road) or a liquid with a concentration gradient (e.g. sugar solution where the concentration decreases from top to bottom).
- Many crystals are anisotropic, i.e. the propagation speed depends on the direction (relative to the crystal layers). This leads to phenomena like *birefringence*.

REFRACTIVE INDEX

The propagation speed in a homogeneous and isotropic medium can be written as

$$c_M = \frac{c}{n}$$

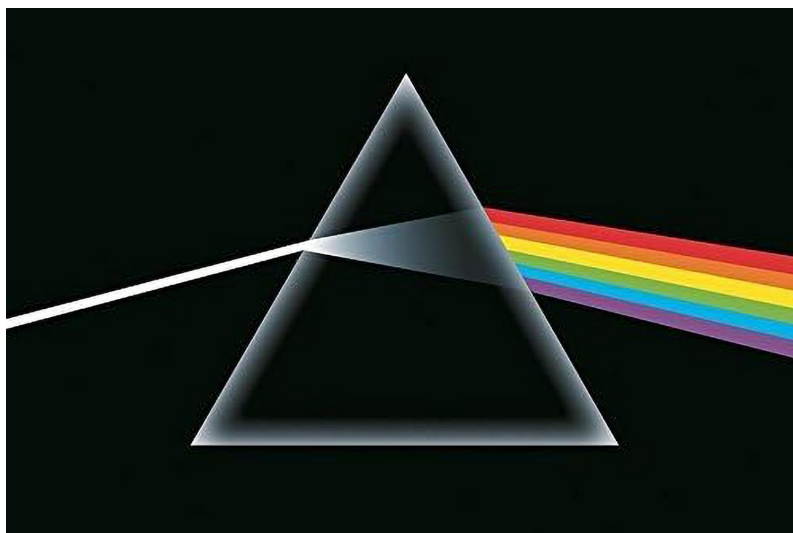
where n is the refractive index of the medium*.

REMARKS:

- For homogeneous and isotropic media, rays are only bent when light goes from a first into a second medium with a different refractive index.
- A medium with a high refractive index is said to have a high *optical density*.

DISPERSION

The propagation speed in a medium (and therefore the refractive index) depends on the wavelength. This is known as *dispersion*. A well-known example of dispersion is that white light can be split up in the underlying colour spectrum using a glass prism.

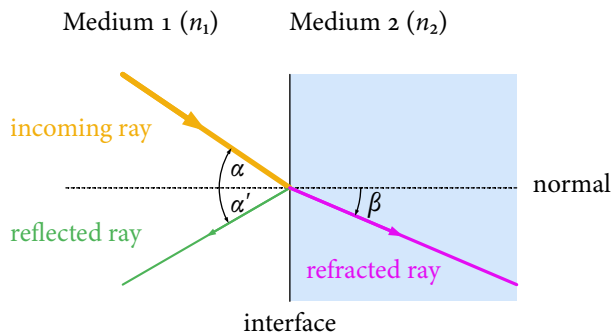


Cover image of Pink Floyd's iconic album *The Dark Side of the Moon*

* See *Formulae, Tables, Concepts*, p 193, for a table listing values for some common media.

REFLECTION AND REFRACTION

When light passes from a medium with refractive index n_1 to another medium with refractive index n_2 , it can be *reflected* or *refracted*[†]:



The relations between the *angle of incidence* α , the *angle of reflection* α' , and the *angle of refraction* β are given by the *law of reflection*

$$\alpha' = \alpha$$

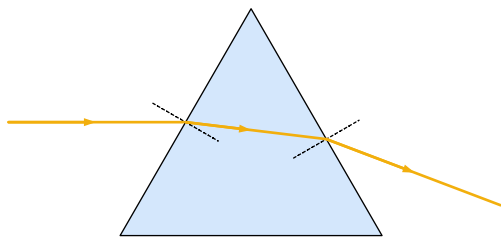
and the *law of refraction* (Snell's law)

$$n_1 \cdot \sin \alpha = n_2 \cdot \sin \beta$$

The deflection for light going from a medium with a low refractive index to a medium with a high refractive index (e.g. from air to glass or water) is always *towards the normal*, while it is away from the normal in the opposite case.

EXAMPLE: Path of a light ray in a triangular prism

When the light ray enters the prism, it is deflected towards the normal. When it leaves the prism, it is bent away from the normal.

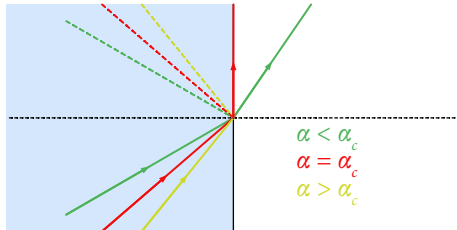


[†] In general, there is both a reflected and a refracted wave. The intensity of the incoming wave is split up between the reflected and refracted wave. In the case of *total internal reflection* (see later), there is only a reflected wave with the full intensity.

TOTAL INTERNAL REFLECTION

For light rays going from a medium with a high refractive index into a medium with a low refractive index, there is an upper limit for the angle of incidence where refraction is possible:

Medium 1 (n_1) Medium 2 ($n_2 < n_1$)



The *critical angle* α_c is defined by the condition

$$n_1 \cdot \sin \alpha_c = n_2 \cdot \sin \frac{\pi}{2} = n_2 \quad \Rightarrow \quad \sin \alpha_c = \frac{n_2}{n_1}$$

For $\alpha < \alpha_c$ there is both a reflected and a refracted ray, but at $\alpha = \alpha_c$ the refracted ray disappears, and for $\alpha \geq \alpha_c$ only the reflected ray can be observed. The transition is continuous, i.e. the intensity of the refracted ray decreases and the intensity of the reflected ray increases while the angle of incidence increases.

The phenomenon that there is only reflection for angles of incidence greater than the critical angle is known as *total internal reflection*.

EXAMPLE: Porro prism

A *Porro prism* (named after its inventor Ignazio Porro) is used to change the orientation of an image. It consists of a right triangle prism. When a light ray enters the prism through its base, it is twice internally reflected (TR) before it leaves the prism back through the base. The outgoing ray is perpendicular to the incoming ray. This is obvious for light rays perpendicular to the base, but it is also true for arbitrary angles of incidence.

